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Event-triggered reinforcement learning-based safe control for stochastic systems subject to asymmetric input constraints and unknown dynamics

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ABSTRACT

This paper investigates the safe optimal control (SOC) for input-constrained unknown stochastic systems via adaptive dynamic programming (ADP) and generalized fuzzy hyperbolic model (GFHM). Firstly, a GFHM is employed to approximate the unknown nonlinear terms of the stochastic system, thereby eliminating the need for exact system modeling. Secondly, a modified nonquadratic value function is developed to address the problem of asymmetric input constraints (AICs), where a control barrier function (CBF) with a damping factor is incorporated to penalize unsafe behaviors. Thirdly, based on ADP, the optimal control policy is obtained by solving the modified Hamilton-Jacobi-Bellman (HJB) equation. Furthermore, a dynamic event-triggered mechanism (DETM) is proposed to alleviate the computational burden and conserve communication resources. In addition, the Lyapunov method is utilized to guarantee the uniform ultimate boundedness (UUB) of the system states and the weight estimation errors of the critic neural networks (CNNs). Finally, two simulation examples are provided to demonstrate the effectiveness of the proposed control strategy.

1. Introduction

Stochastic systems are widely present in practical scenarios for example ecological management (Zhang et al., 2025b), financial decision-making (Syed et al., 2025), and engineering control (Zhu & Huang, 2021). The evolutionary processes are influenced by random perturbations, uncertain inputs, or stochastic parameters, making it difficult to directly apply traditional deterministic control methods. Stochastic optimal control (OC) involves two core challenges: data-based uncertainty modeling of unknown dynamics, and approximate solution of the analytically intractable stochastic HJB equation in Wei et al. (2023) and Si et al. (2024). Although Si et al. (2024) and Ming et al. (2022) studied OC for stochastic systems, they did not consider the safety requirements of the system.

A critical challenge in stochastic control designs the effectively address the problem of unknown nonlinear dynamics. In recent research, to solve the problem of nonlinearities and uncertainties, various approximation methods have been developed, including neural networks (NNs) (Ma et al., 2026), generalized fuzzy hyperbolic model (GFHM) (Zhang et al., 2017) and fuzzy logic systems (FLSs) (Zhang et al., 2024). Among

these approaches, GFHM demonstrates remarkable advantages by utilizing hyperbolic tangent functions as basis functions, which circumvents the basis function selection challenge in conventional NNs and provides rapid, accurate estimation without requiring precise system modeling (Liu et al., 2017; Qin et al., 2026b; Zhang et al., 2019). Recent studies have successfully applied GFHM to self-triggered control for model-free systems (Ming et al., 2023), on secure OC of nonlinear systems (Wang et al., 2025), and OC of input-constrained stochastic systems (Zhang et al., 2025d). However, the GFHM established above is proposed for single-input systems, and its application research in multi-input stochastic systems remains unexplored.

On the other hand, safety has become a critical issue in modern engineering systems (Wang et al., 2023), covering a wide range of fields such as multi-agent coordination (Zhang et al., 2025c), robotic navigation (Wang et al., 2022), unmanned aerial vehicle (UAV) obstacle avoidance (Dai et al., 2020) and autonomous driving (Lin et al., 2023b). In these applications, any violation of safety constraints-such as collisions, speeding, exceeding operating limits can cause irreversible damage to equipment, serious environmental hazards, and even personal injury. Therefore, designing control strategies that can simultaneously ensure

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system security and achieve expected performance goals has become a challenging research frontier in the field of control. In [Jiang and Guo \(2023\)](#), Jiang et al. developed CBF-enhanced model predictive control for human-robot collaborative workspaces, maintaining minimum separation distances while optimizing task performance. In [Li et al. \(2025b\)](#), Li et al. integrated CBFs into motion planning to enforce collision avoidance and lane-keeping constraints, achieving safe navigation through complex traffic scenarios. In [Habibi et al. \(2023\)](#), Habibi et al. designed a hierarchical control structure utilizing barrier Lyapunov functions (BLFs) for quadrotors to simultaneously address obstacle avoidance and trajectory tracking, achieving safe autonomous flight under uncertain dynamics and environmental disturbances. Similarly, Qin et al. proposed a consensus protocol for multi-agent robot teams based on distributed CBF to address the non-zero-sum (NZS) differential game problem of nonlinear systems, ensuring formation safety and avoiding collisions between intelligent agents in [Qin et al. \(2024a\)](#). Despite these advances, most existing CBFs are limited to deterministic systems, and the safety issue in stochastic systems has not yet been effectively addressed.

Additionally, input limited control is a special scenario of constrained control, all actual actuators (robotic manipulators, aircraft, and chemical processes) have input range limitations ([Ma et al., 2026](#); [Qin et al., 2026b](#)), if the control algorithm does not consider input limitation, it may lead to system control performance degradation ([Habibi et al., 2023](#); [Wang et al., 2022](#)), thus the issue of input constraints has received widespread attention. In [Yao et al. \(2024\)](#), the scholars studied the use of flexible performance control methods for systems with limited inputs and unmeasurable states. In [Wu and X. Li \(2022\)](#), the authors designed a synchronization control strategy for complex dynamic networks under input saturation based on finite-time theory. However, actual actuators often exhibit asymmetric positive and negative saturation boundaries, necessitating different treatment methods. In [Yao et al. \(2025a\)](#), dual flexible performance control was studied for high-order constrained system. In [Song et al. \(2024\)](#), Song et al. utilized a Nussbaum function and a time-varying BLF to address asymmetric saturation and dead zone issues. In [Yao et al. \(2025b\)](#), Yao et al. developed a control scheme guided by a sliding flexible performance boundary for nonlinear systems with input constraints. Nevertheless, the aforementioned approaches focus on deterministic nonlinear systems without accounting for stochastic uncertainties. Although the aforementioned methods effectively resolved input constraint issues, they mostly focus on feasibility and stability analysis, with insufficient consideration for optimality.

ADP can approximate the solution of HJB equation online through algorithms, such as value iteration or strategy iteration, combined with function approximation techniques (such as NNs), in order to obtain a near OC strategy without needing to know the complete system dynamics ([Qin et al., 2026a, 2025a](#); [Zhang et al., 2025a](#)). In [Guo et al. \(2025\)](#), Guo et al. studied the event-triggered OC problem for NZS game systems with AICs. In [Qin et al. \(2024b\)](#), Qin et al. studied the secure OC problem for NZS game systems based on ADP. For practical applications, the ADP-based OC method was investigated for multi-agent systems (MASs) in [Zhang et al. \(2026a\)](#). Lin et al. studied the OC problem of massive multi-agent systems based on ADP in [Lin et al. \(2026\)](#). Despite these advances, research on OC problems with AICs for stochastic systems is limited, which requires further research.

To reduce the communication and computational burdens caused by unnecessary sampling, the event-triggered ADP methods have been widely investigated within the domain of OC ([Hu et al., 2026](#); [Liu et al., 2025](#); [Pan et al., 2024](#); [Wei & Jiang, 2025](#)). Different from static event-triggered mechanisms (ETM), dynamic event-triggering mechanisms (DETM) can generate larger inter-event intervals by incorporating internal dynamic variables, thereby further reducing communication frequency ([Qin et al., 2025b](#); [Zhang et al., 2026b](#)). In [Lin et al. \(2023a\)](#), the DETM was applied to multiplayer Stackelberg-Nash games system, which effectively reduce communication burdens. In [Li et al. \(2025a\)](#),

a distributed prescribed-time controller based on DETM was developed for heterogeneous multi-agent systems. These studies demonstrate that DETM can effectively reduce system resource consumption while maintaining control performance.

According to the aforementioned research, this paper is devoted to the safe OC issue of nonlinear stochastic systems with AICs. The main innovations are as follows:

- 1) This paper develops a safety OC framework for uncertain nonlinear stochastic systems under asymmetric actuation limits, in which the ADP approach is used, and the stochastic OC strategies can be obtained by solving the improved HJB equation embedding a CBF.
- 2) A GFHM-based identifier is designed for multi-input stochastic systems, which approximates unknown nonlinear dynamics using hyperbolic tangent basis functions without requiring precise system modeling. Rigorous analysis confirms that the identification error satisfies the mean-square uniformly ultimately bounded (UUB).
- 3) Unlike the proposed methods in [Li et al. \(2025b\)](#), [Habibi et al. \(2023\)](#), [Qin et al. \(2024a\)](#), the CBF-augmented nonquadratic cost function is constructed for stochastic systems to simultaneously handle safety constraints and asymmetric input limitations issues. In addition, to balance safety and optimal performance, a damping coefficient is introduced.
- 4) Compared with static event-triggered mechanism ([Hu et al., 2026](#); [Wei et al., 2023](#); [Zhu & Huang, 2021](#)), DETM is developed to adaptively adjust triggering thresholds via introducing internal dynamic variables in the triggering conditions, which generates larger inter-event intervals in contrast to static ETM, thereby effectively reducing computational burden and communication resources.

The organization of this paper is shown as follows: [Section 2](#) formulates the problem and introduces the system model. [Section 3](#) presents the GFHM-based identification. In [Section 4](#), safe and robust OC design. In [Section 5](#), designs the dynamic event-triggered controls and critic NNs. In [Section 6](#), two examples and simulation results are presented. Finally, the conclusion is shown in [Section 7](#).

2. Problem formulation

Consider the following nonlinear stochastic system as

$$\begin{aligned} d\zeta &= F(\zeta, \mu_j)dt + h(\zeta)dw \\ &= \left[f(\zeta) + \sum_{j=1}^N E_j \mu_j \right] dt + h(\zeta)dw, \end{aligned} \quad (1)$$

where $\zeta(t) \in \mathbb{R}^n$ represents the system state, $\mu_j \in \mathbb{R}^n$ is the control law of the j th controller and $w(t)$ is the random variable which has its own probability distribution. The unknown nonlinear function $f(\zeta) = [f_1(\zeta), \dots, f_n(\zeta)]^T$ satisfies $f(0) = 0$. $E_j \in \mathbb{R}^{n \times n}$, $h(\cdot) \in \mathbb{R}^{n \times n}$ denote Lipschitz continuous function and $h(\cdot) \leq \bar{h}$, $h(0) = 0$. $F(\zeta, \mu_j) = f(\zeta(t)) + \sum_{j=1}^N E_j \mu_j(t)$ is local Lipschitz continuous.

Assume that the control inputs are restricted as

$$\mu_{j,\min} \leq \mu_{j,l} \leq \mu_{j,\max}, \quad (2)$$

where $\mu_{j,l}$ denotes the l th control signal of the j th controller with $l \in 1, 2, \dots, n$. $\mu_{j,\min}$, $\mu_{j,\max}$ represent the asymmetric saturation limits of the controller, $|\mu_{j,\min}| \neq |\mu_{j,\max}|$.

Definition 1 ([Zhang et al. \(2025d\)](#)). Consider the stochastic nonlinear system (1) and define $\mathcal{A}$ is a differential symbol, which is given as

$$\mathcal{A}\mathcal{V}(\zeta) = \frac{\partial \mathcal{V}}{\partial \zeta} F(\zeta, \mu_j) + \frac{1}{2} \text{Tr} \left\{ h^T(\zeta) \frac{\partial^2 \mathcal{V}}{\partial \zeta^2} h(\zeta) \right\}. \quad (3)$$

3. Design GFHM identifier

To cope with the system uncertainty, a GFHM-based identifier is constructed, which benefits from the universal approximation property of

fuzzy hyperbolic models. Initially, the generalized fuzzy hyperbolic rule (GFHR) base is subsequently defined.

Definition 2. The GFHR base is defined as a fuzzy system comprising 2^M rules that through transformation-based positive and negative fuzzy sets, these rules map n input variables to the system output.

Lemma 1 (Zhang et al. (2019)). According to Definition 2, for a multiple-input single output system, the membership functions of v_{N_q} and v_{N_q} are expressed as

$$v_{P_q}(\bar{c}_q) = e^{-\frac{1}{2}(\bar{c}_q - \kappa_q)^2} \quad (4)$$

$$v_{N_q}(\bar{c}_q) = e^{-\frac{1}{2}(\bar{c}_q - \kappa_q)^2} \quad (5)$$

with $\kappa_q \in \mathbb{R}^+$, $\bar{c}_q = \zeta_i - c_{ij}$, $q = m_1 + \dots + m_{i-1}$, $c_{ij} \in \mathbb{R}^+$ being the bias parameter for the j th transformation, $m_i \in \mathbb{Z}^+$ being the number of transformations related to the ζ_i .

Subsequently, we can be derived that:

$$\begin{aligned} y &= \sum_{q=1}^M \frac{c_{P_q} e^{l_q \bar{c}_q} + c_{N_q} e^{-l_q \bar{c}_q}}{e^{l_q \bar{c}_q} + e^{-l_q \bar{c}_q}} \\ &= \sum_{q=1}^M a_q + \sum_{q=1}^M b_q \frac{e^{l_q \bar{c}_q} + e^{-l_q \bar{c}_q}}{e^{l_q \bar{c}_q} + e^{-l_q \bar{c}_q}} \\ &= I + \Xi^T \tanh(l_G \bar{\zeta}) \end{aligned} \quad (6)$$

where $I = \sum_{q=1}^M a_q$, $a_q = \frac{c_{P_q} + c_{N_q}}{2}$ signifies bias term and $\Xi = [b_1, \dots, b_M]^T$, $b_q = \frac{c_{P_q} - c_{N_q}}{2}$, $l_G = \text{diag}[l_1, \dots, l_M]$, and $\tanh(l_G \bar{\zeta}) = [\tanh(l_1 \bar{\zeta}_1), \dots, \tanh(l_M \bar{\zeta}_M)]^T$. Hence, (6) constitutes a GFHM.

Lemma 2 (Wang et al. (2025)). On the compact domain C^1 , any continuous function $g(\zeta)$ can be approximated by a function $G(\zeta)$ from the GFHM class with any positive tolerance π , as follows:

$$\sup_{\zeta \in \Theta} |g(\zeta) - G(\zeta)| < \pi$$

with $\pi \in \mathbb{R}^+$. That is, the GFHM possesses universal approximation capability.

A theorem related to GFHM is presented in the following presentation, in the following theorem, the unknown system dynamics and stochastic-induced nonlinearities are collectively represented by the composite function $g = f(\zeta) + \frac{3}{4}\|h\|^4 \zeta$. Leveraging the approximation capability of GFHM, this composite function g is given as

$$g = I + \Xi^T \tanh(l(X - C)) + \epsilon \quad (7)$$

with $I = [I_1, I_2, \dots, I_n]^T \in \mathbb{R}^n$, $\Xi = [\Xi_1^T, \Xi_2^T, \dots, \Xi_n^T]^T \in \mathbb{R}^{n \times kn^2}$, $C = [C_1^T, C_2^T, \dots, C_n^T]^T \in \mathbb{R}^{kn^2}$ being ideal weights, which are unknown. ϵ is approximation error and $\|\epsilon\| \leq \epsilon_s, \epsilon_s \in \mathbb{R}^+$.

Then, GFHM is established as follows

$$\dot{\hat{\zeta}} = \hat{I} + \hat{\Xi}^T \tanh(l(\hat{X}_s - \hat{C})) + D\zeta + \sum_{j=1}^N \hat{E}_j \mu_j. \quad (8)$$

where the stable design matrix $D = [d_1, d_2, \dots, d_n] \in \mathbb{R}^{n \times n}$, $\hat{I}, \hat{\Xi}, \hat{C}, \hat{E}_j$ are estimated value of ideal parameter matrices $I, \Xi, C, E_j = [E_{j1}, E_{j2}, \dots, E_{jn}]^T \in \mathbb{R}^n$, define $\tilde{I} = I - \hat{I}$, $\tilde{\Xi} = \Xi - \hat{\Xi}$, $\tilde{C} = C - \hat{C}$, $\tilde{E}_j = E_j - \hat{E}_j$. The augmented state vector $X = \Phi(X_s) = [\underbrace{X_s^T, \dots, X_s^T}_{n}, \dots]^T \in \mathbb{R}^{kn^2}$, and $\hat{X} \triangleq \Phi(\hat{X}_s) = \Phi(\zeta \otimes I_k)$.

Meanwhile, inspired by Qin et al. (2026b), we choose $l = 1$.

Letting $\zeta = \zeta - \hat{\zeta}$, invoking (1) and (8), the dynamics of the identifier error is

$$d\zeta = \left\{ f - \hat{I} - \hat{\Xi}^T \tanh(\hat{X} - \hat{C}) - D\zeta + \sum_{j=1}^N \tilde{E}_j \mu_j \right\} dt + h dw \quad (9)$$

Assumption 1. $\tilde{I}, \tilde{\Xi}, \tilde{C}, \tilde{\Xi}$ are positive numbers. For the ideal parameter matrices, the following bound holds:

- 1) $\|\Xi\| \leq \tilde{\Xi}$
- 2) $\|I\|_F \leq \tilde{I}, \|\Xi\|_F \leq \tilde{\Xi}, \|C\|_F \leq \tilde{C}$.

Theorem 1. For system (1) and Assumption 1, the GFHM-based identifier (8) along with the update laws (10)–(12) ensures that all error signals are UUB in the mean square sense.

$$\dot{\hat{I}} = \alpha_1 \|\zeta\|^2 \zeta - \beta_1 \alpha_1 \hat{I} \quad (10)$$

$$\dot{\hat{\Xi}} = \alpha_{2,v} \|\zeta_v\|^2 \zeta_v \tanh(\hat{X}_s - \hat{C}_v) - \beta_2 \alpha_{2,v} \hat{\Xi}_v \quad (11)$$

$$\dot{\hat{C}} = -\alpha_3 \beta_3 \hat{C}, \dot{\hat{E}}_j = \alpha_{4,j} \|\zeta\|^2 \zeta \mu_j^T \quad (12)$$

where $\beta_i \in \mathbb{R}^+, i = 1, 2, 3$, $\alpha_1, \alpha_3 \in \mathbb{R}^+$ are learning rates, $\alpha_{4,j} \in \mathbb{R}^{n \times n}$ denote symmetric positive definite matrices and $\alpha_{2,v} \in \mathbb{R}^{kn \times kn}$, $v = 1, 2, \dots, n$.

Proof. The candidate Lyapunov function is given as

$$\mathcal{V} = \sum_{j=1}^5 \mathcal{V}_j, j = 1, 2, \dots, 5 \quad (13)$$

where

$$\mathcal{V}_1 = \frac{1}{4k^2 n_1^2} (\tilde{X}^T \tilde{X})^2 = \frac{1}{4} (\zeta^T \zeta)^2, \mathcal{V}_2 = \frac{1}{2} \text{Tr} \{ \tilde{I}^T \alpha_1^{-1} \tilde{I} \}$$

$$\mathcal{V}_3 = \frac{1}{2} \sum_{v=1}^n \text{Tr} \{ \tilde{\Xi}_v^T \alpha_{2,v}^{-1} \tilde{\Xi}_v \}, \mathcal{V}_4 = \frac{1}{2} \text{Tr} \{ \tilde{C}^T \alpha_3^{-1} \tilde{C} \}$$

$$\mathcal{V}_5 = \frac{1}{2} \sum_{j=1}^N \text{Tr} \{ \tilde{E}_j^T \alpha_{4,j}^{-1} \tilde{E}_j \}.$$

By applying the Cauchy-Schwartz and Young's inequalities, it can get

$$\begin{aligned} \mathcal{A}\mathcal{V}_1 &= \|\zeta\|^2 \zeta^T \left\{ f - \hat{I} - \hat{\Xi}^T \tanh(\hat{X} - \hat{C}) - D\zeta + \sum_{j=1}^N \tilde{E}_j \mu_j \right\} \\ &\quad + \frac{3}{2} \text{Tr} \{ h \zeta \zeta^T h \} \\ &\leq \|\zeta\|^2 \zeta^T \left\{ f - \hat{I} - \hat{\Xi}^T \tanh(\hat{X} - \hat{C}) - D\zeta + \sum_{j=1}^N \tilde{E}_j \mu_j \right. \\ &\quad \left. + \frac{3}{4} \|h\|^4 \zeta \right\} + \frac{3}{4} \\ &= \|\zeta\|^2 \zeta^T \left\{ -\hat{I} - \hat{\Xi}^T \tanh(\hat{X} - \hat{C}) - D\zeta + \sum_{j=1}^N \tilde{E}_j \mu_j + I \right. \\ &\quad \left. + \Xi^T \tanh(X - C) + \epsilon \right\} + \frac{3}{4} \\ &= \|\zeta\|^2 \zeta^T \left\{ \tilde{I} + \Xi^T [\tanh(X - C) - \tanh(\hat{X} - \hat{C})] - D\zeta \right. \\ &\quad \left. + \sum_{j=1}^N \tilde{E}_j \mu_j + \tilde{\Xi}^T \tanh(\hat{X} - \hat{C}) + \epsilon \right\} + \frac{3}{4} \\ &\leq \|\zeta\|^2 \zeta^T \tilde{I} + 2nk^{\frac{1}{2}} \|\zeta\|^2 \zeta^T \tilde{\Xi} - \|\zeta\|^2 \zeta^T D\zeta + \|\zeta\|^2 \zeta^T \tilde{\Xi}^T \tanh(\hat{X} - \hat{C}) \\ &\quad + \|\zeta\|^2 \zeta^T \sum_{j=1}^N \tilde{E}_j \mu_j + \|\zeta\|^2 \zeta^T \epsilon + \frac{3}{4} \\ &\leq \|\zeta\|^2 \zeta^T \tilde{I} + 2nk^{\frac{1}{2}} \|\zeta\|^2 \zeta^T \tilde{\Xi} + \lambda_{\max}(-D) \|\zeta\|^4 + \frac{3}{4} \|\zeta\|^4 + \frac{1}{4} \epsilon_s^4 \\ &\quad + \|\zeta\|^2 \zeta^T \tilde{\Xi}^T \tanh(\hat{X} - \hat{C}) + \|\zeta\|^2 \zeta^T \sum_{j=1}^N \tilde{E}_j \mu_j + \frac{3}{4} \end{aligned} \quad (14)$$

where $\|\tanh(X - C) - \tanh(\hat{X} - \hat{C})\| \leq 2nk^{\frac{1}{2}}$.

$$\begin{aligned} \mathcal{A}\mathcal{V}_2 &= \text{Tr} \{ \tilde{I}^T \alpha_1^{-1} [-\alpha_1 \|\zeta\|^2 \zeta + \beta_1 \alpha_1 (I - \tilde{I})] \} \\ &\leq -\|\zeta\|^2 \zeta^T \tilde{I} + \beta_1 \tilde{I} \|\tilde{I}\|_F - \beta_1 \|\tilde{I}\|_F^2 \end{aligned} \quad (15)$$

$$\begin{aligned} \mathcal{A}\mathcal{Y}_3 &= \sum_{v=1}^n \text{Tr} \left\{ \tilde{\alpha}_v^{-1} \left[-\alpha_{2,v} \|\tilde{\zeta}_v\|^2 \tilde{\zeta}_v \tanh(\hat{X}_s - \hat{C}_v) \right. \right. \\ &\quad \left. \left. + \beta_2 \alpha_{2,v} (\Xi_v - \tilde{\Xi}_v) \right] \right\} \\ &\leq -\|\tilde{\zeta}\|^2 \tilde{\zeta}^T \tilde{\Xi} \tanh(\hat{X} - \hat{C}) + \beta_2 \tilde{\Xi} \|\tilde{\Xi}\|_F - \beta_2 \|\tilde{\Xi}\|_F^2 \end{aligned} \quad (16)$$

$$\begin{aligned} \mathcal{A}\mathcal{Y}_4 &= \text{Tr} \left\{ \tilde{C}^T \alpha_3^{-1} [\beta_3 \alpha_3 (C - \tilde{C})] \right\} \\ &\leq \beta_3 \tilde{C} \|C\|_F - \beta_3 \|\tilde{C}\|_F^2 \end{aligned} \quad (17)$$

$$\begin{aligned} \mathcal{A}\mathcal{Y}_5 &= \sum_{j=1}^N \text{Tr} \left\{ \tilde{E}_j^T \alpha_{4,j}^{-1} \left(-\alpha_{4,j} \|\tilde{\zeta}\|^2 \tilde{\zeta} \mu_j^T \right) \right\} \\ &= -\sum_{j=1}^N \|\tilde{\zeta}\|^2 \tilde{\zeta}^T \tilde{E}_j \mu_j \end{aligned} \quad (18)$$

Considering (14)–(18) and Assumption 1, it can get

$$\begin{aligned} \mathcal{A}\mathcal{Y} &\leq -\left(\lambda_{\max}(-D) - \frac{3}{4} - \frac{3}{2}nk^{\frac{1}{2}} \right) \|\tilde{\zeta}\|^4 - \beta_1 \left(\|\tilde{I}\|_F - \frac{1}{2}\tilde{I} \right)^2 \\ &\quad - \beta_2 \left(\|\tilde{\Xi}\|_F - \frac{1}{2}\tilde{\Xi} \right)^2 - \beta_3 \left(\|\tilde{C}\|_F - \frac{1}{2}\tilde{C} \right)^2 + \Upsilon \end{aligned} \quad (19)$$

where $\Upsilon = \frac{1}{2}nk^{\frac{1}{2}}\tilde{\Xi}^4 + \frac{1}{4}\epsilon_s^4 + \frac{1}{4}\beta_1\tilde{I}^2 + \frac{1}{4}\beta_2\tilde{\Xi}^2 + \frac{1}{4}\beta_3\tilde{C}^2 + \frac{3}{4}$,

$\lambda_{\max}(-D) - \frac{3}{4} - \frac{3}{2}nk^{\frac{1}{2}} > 0$, which $D < 0$.

Based on (19) and (13), it obtains

$$\mathcal{A}\mathcal{Y} \leq -\gamma\mathcal{Y} + \Upsilon \quad (20)$$

with $\gamma = \min \left\{ 4\lambda_{\max}(-D) - 3 - 6nk^{\frac{1}{2}}, \beta_b \right\}$, $b = 1, 2, 3$.

According to Zhang et al. (2026b, Lemma 1), it yields that

$$\mathbb{E}\{\mathcal{Y}\} \leq \exp(-\gamma t)\mathcal{Y}(0) + \frac{\Upsilon}{\gamma}. \quad (21)$$

Hence, all errors are mean-square bounded. The proof is completed. $\square$

Remark 1. In contrast to traditional NNs, the GFHM, as a universal approximator, features linearly increasing parameters with input dimensions and a fixed structure that obviates cumbersome trial-and-error configuration. The bounded derivatives of its basis functions simplify Lyapunov stability analysis. Retaining fuzzy inference semantics, GFHM is well suited for adaptive and OC of nonlinear systems.

Remark 2. The following results are instrumental for the controller design. Initially, the GFHM undergoes a training phase to identify system dynamics. Subsequently, the reconstructed model (8) obtained through the GFHM identifier serves as the foundation for both the control law derivation and the critic NN weight tuning. Nevertheless, throughout the entire process of theoretical development, we have retained the original system (1) for further analysis.

4. Safe and robust OC design

In this section, a nonquadratic cost function is constructed to formulate the safe and robust OC problem with handling the problem of AICs. The operational constraints and safety conditions constitute set $\mathfrak{D}$ of a safety-critical system (1) is

$$\mathfrak{D} = \{\zeta \mid S(\zeta) \geq 0\} \quad (22)$$

where $S(\zeta)$ is a smooth function with well-defined derivatives.

Here, $\text{int } \mathfrak{D}$ represents its interior, the boundary of $\mathfrak{D}$ is denoted by $\partial\mathfrak{D}$. The safe set comprises three regions:

$$\begin{cases} \mathfrak{D} = \{\zeta \in \mathbb{R}^n \mid S(\zeta) \geq 0\}, \\ \partial\mathfrak{D} = \{\zeta \in \mathbb{R}^n \mid S(\zeta) = 0\}, \\ \text{int}(\mathfrak{D}) = \{\zeta \in \mathbb{R}^n \mid S(\zeta) > 0\}, \end{cases} \quad (23)$$

Assumption 2. To enforce safety, a CBF $B_r(\zeta)$ is monotonically decreasing and satisfies the following properties:

1. $B_r(\zeta) \geq 0, \forall \zeta \in \text{int}(\mathfrak{D})$,
2. $B_r(\zeta) \rightarrow \infty, \forall \zeta \rightarrow \partial\mathfrak{D}$.

The CBF is formulated as:

$$B_r(\zeta) = \rho \frac{\mathfrak{s}(\zeta)}{S(\zeta)} \quad (24)$$

where $\rho > 0$ represents the damping coefficient, which is used to balance the safety and performance of the system, $B_r(\zeta)$ can be selected via determining the special scheduling function $\mathfrak{s}(\zeta)$ by the user.

Remark 3. The selection of $\mathfrak{s}(\zeta)$ needs to meet the following characteristics: $\mathfrak{s}(\zeta)$ is within the safe set; when ζ approaches the boundary, $\mathfrak{s}(\zeta)$ does not cancel out the tendency of the denominator $S(\zeta)$ to approach zero. In Example 1, when the system state is within the interior, $S(\zeta) > 0$ and $\mathfrak{s}(\zeta) > 0$, so $B_r(\zeta) \geq 0$; as the system state nears the region boundary, $S(\zeta)$ tends to zero, and $\mathfrak{s}(\zeta)$ tends to $\frac{\pi}{2}$, so $B_r(\zeta) \rightarrow \infty$. Overall, Assumption 2 is met.

Select the appropriate function $B_r(\zeta)$ based on the above discussion, define the i th value function is

$$\mathcal{V}_i(\zeta(t)) = \mathbb{E} \left\{ \int_t^\infty e^{-a(s-t)} (\zeta^T Q_i \zeta + \sum_{j=1}^N \mathcal{R}_i(\mu_j) + B_r(\zeta)) ds \mid \mathcal{F}_t \right\} \quad (25)$$

and

$$\mathcal{R}_i(\mu_j) = 2\rho_{1,j} \int_{\rho_{2,j}}^{\mu_j} \phi^{-\top} \left(\frac{t - \rho_{2,j}}{\rho_{1,j}} \right) R_{ij} dt \quad (26)$$

where $R_{ij} = \text{diag} \left(r_{ij}^1, \dots, r_{ij}^n \right) > 0$, $\phi(\cdot)$ is chosen to be the hyperbolic tangent function $\tanh(\cdot)$, which is a monotonic and bounded odd function, and $\phi(0) = 0$. $\rho_{1,j} = \frac{\mu_{j,\max} - \mu_{j,\min}}{2}$, $\rho_{2,j} = \frac{\mu_{j,\max} + \mu_{j,\min}}{2}$.

Then, the optimal value function is

$$\begin{aligned} \mathcal{V}_i(\zeta(t))^* &= \min_{\mu_i} \mathbb{E} \left\{ \int_t^\infty e^{-a(s-t)} \left(\zeta^T Q_i \zeta + \sum_{j=1}^N \mathcal{R}_i(\mu_j) + B_r(\zeta) \right) ds \mid \mathcal{F}_t \right\} \end{aligned} \quad (27)$$

For the multi-player NZS differential game, the objective is to determine an admissible Nash equilibrium policy $\{\mu_1^*, \dots, \mu_N^*\}$, as characterized in Definition 3 below.

Definition 3. For the NZS game associated with system (1), an admissible control policies $\{\mu_1^*, \dots, \mu_N^*\}$, $\forall i \in N$, yields a Nash equilibrium when it satisfies the subsequent inequalities

$$\begin{aligned} V_i^* &\triangleq V_i(\zeta, \mu_1^*, \dots, \mu_i^*, \dots, \mu_N^*) \\ &\leq V_i(\zeta, \mu_1^*, \dots, \mu_i, \dots, \mu_N^*) \end{aligned} \quad (28)$$

Given control strategy $\{\mu_1, \dots, \mu_N\}$ and a continuously differentiable value function $\mathcal{V}_i(\zeta)$ on the compact set Ω .

Based on (25), we have

$$\begin{aligned} \nabla \mathcal{V}_i^T(\zeta) [f + \sum_{j=1}^N E_j \mu_j] - a\mathcal{V}_i(\zeta) + \frac{1}{2} \text{Tr} \{ h^T \nabla^2 \mathcal{V}_i h \} \\ + \zeta^T Q_i \zeta + \sum_{j=1}^N \mathcal{R}_i(\mu_j) + B_r(\zeta) = 0 \end{aligned} \quad (29)$$

with $\nabla \mathcal{V}_i \in \mathbb{R}^n$ being the gradient of $\mathcal{V}_i$ with respect to ζ , $\mathcal{V}_i(0) = 0$.

Then, the Hamiltonian function is

$$\begin{aligned} H_i(\zeta, \mu_1, \dots, \mu_N, \mathcal{V}_i(\zeta)) &= \nabla \mathcal{V}_i^T(\zeta) F(\zeta, \mu_i) - a\mathcal{V}_i(\zeta) + \sum_{j=1}^N \mathcal{R}_i(\mu_j) \\ &\quad + \frac{1}{2} \text{Tr} \{ h^T \nabla^2 \mathcal{V}_i h \} + \zeta^T Q_i \zeta + B_r(\zeta) \end{aligned} \quad (30)$$

where $F(\zeta, \mu_i) = f(\zeta(t)) + \sum_{j=1}^N E_j \mu_j(\zeta(t))$, $\mathcal{L}\mathcal{V}_i = \mathcal{V}_i^T F(\zeta, \mu_i) + \frac{1}{2} \text{Tr} \{ h^T \nabla^2 \mathcal{V}_i h \}$ and $\nabla \mathcal{V}_i = \frac{\partial \mathcal{V}_i}{\partial \zeta}$, $\nabla^2 \mathcal{V}_i = \frac{\partial^2 \mathcal{V}_i}{\partial \zeta^2}$, $i = 1, \dots, n$.

From the necessary condition for optimality, $\frac{\partial H_i}{\partial \mu_i} = 0$, the OC policy is

$$\mu_i^* = \rho_{1,i} \tanh \left(-\frac{1}{2\rho_{1,i}} R_{ii}^{-1} E_i^T \nabla \mathcal{V}_i^*(\zeta) \right) + \rho_{2,i} \quad (31)$$

Substituting (27) and (31) into (30), it follows that

$$\begin{aligned} H_i(\zeta, \mu_1^*, \dots, \mu_N^*, \mathcal{V}_i^*(\zeta)) &= \nabla \mathcal{V}_i^{*T}(\zeta) [f + \sum_{j=1}^N E_j \mu_j^*] - a \mathcal{V}_i^*(\zeta) + \zeta^T Q_i \zeta \\ &\quad + \frac{1}{2} \text{Tr} \{ h^T \nabla^2 \mathcal{V}_i^* h \} + \sum_{j=1}^N \mathcal{R}_i(\mu_j^*) + B_r(\zeta) \quad (32) \\ &= 0 \end{aligned}$$

In what follows, the validity of the cost function (25) and the HJB Eq. (30) is established.

Theorem 2. For system (1), considering the cost function given by (27) and the control strategy presented in (31) guarantees UUB stability under the influence of a standard Wiener process.

Proof. Consider the Lyapunov function

$$\mathcal{Z} = \mathcal{V}_i^* \quad (33)$$

According to Definition 1, we can obtain

$$\mathcal{A} \mathcal{V}_i^*(\zeta) = \nabla \mathcal{V}_i^*(\zeta)^T \left[f(\zeta) + \sum_{j=1}^N E_j \mu_j^*(t) \right] + \frac{1}{2} \text{Tr} \{ h^T \nabla^2 \mathcal{V}_i^* h \} \quad (34)$$

According to (31), we can obtain that

$$\mu_j^* = -\rho_{1,j} \tanh(\Lambda^*) + \rho_{2,j} \quad (35)$$

where $\Lambda^* = \frac{1}{2\rho_{1,j}} R_{ij}^{-1} E_j^T \nabla \mathcal{V}_i^*(\zeta)$.

From (32), it gets

$$\begin{aligned} \nabla \mathcal{V}_i^*(\zeta)^T \left[f + \sum_{j=1}^N E_j \mu_j^* \right] &+ \frac{1}{2} \text{Tr} \{ h^T \nabla^2 \mathcal{V}_i^* h \} \\ &= a \mathcal{V}_i^*(\zeta) - \zeta^T Q_i \zeta - \sum_{j=1}^N \mathcal{R}_i(\mu_j^*) - B_r(\zeta) \quad (36) \end{aligned}$$

Next, perform relevant calculations, we have

$$\begin{aligned} \mathcal{R}_i(\mu_j^*) &= \int_{\rho_{2,j}}^{\mu_j^{*(x)}} 2\rho_{1,j} \tanh^{-1} \left(\frac{t - \rho_{2,j}}{\rho_{1,j}} \right) R_{ij} dt \\ &= 2\rho_{1,j} \tanh^{-1} \left(\frac{\mu_j^* - \rho_{2,j}}{\rho_{1,j}} \right) R_{ij} (\mu_j^* - \rho_{2,j}) \\ &\quad + \rho_{1,j}^2 \bar{R}_{ij}^T \ln \left(\bar{\mathbf{I}} - \left(\frac{\mu_j^* - \rho_{2,j}}{\rho_{1,j}} \right)^2 \right) \\ &= 2\rho_{1,j}^2 \Lambda^{*T} R_{ij} \tanh(\Lambda^*) + \rho_{1,j}^2 \bar{R}_{ij}^T \ln(\bar{\mathbf{I}} - \tanh^2(\Lambda^*)) \quad (37) \end{aligned}$$

where $\bar{R} = [r_1, r_2, \dots, r_n]^T$ and $\bar{\mathbf{I}} = [\underbrace{1, \dots, 1}_n]^T$.

By combining (36) and (37), we can obtain that

$$\begin{aligned} \mathcal{A} \mathcal{V}_i^*(\zeta) &= a \mathcal{V}_i^*(\zeta) - \zeta^T Q_i \zeta - B_r(\zeta) - \sum_{j=1}^N \mathcal{R}_i(\mu_j^*) \\ &\leq -\lambda_{\min}(Q_i) \|\zeta\|^2 - B_r(\zeta) + M \quad (38) \end{aligned}$$

where $M = a \mathcal{V}_i^*(\zeta)$.

Hence, $\mathcal{A} \mathcal{V}_i^*(\zeta) \leq 0$ is ensured whenever ζ satisfies

$$\Upsilon_\zeta = \left\{ \zeta : \|\zeta\| \geq \sqrt{\frac{M}{\lambda_{\min}(Q_i)}} \right\} \quad (39)$$

Consequently, the OC policy guarantees stability of the system (1). Additionally, ζ will converge to a neighborhood with radius $\sqrt{\frac{M}{\lambda_{\min}(Q_i)}}$ near the origin. $\square$

Remark 4. Theorem 2 establishes the theoretical foundation for safe optimal robust control via (31) and (32). However, directly solving the HJB Eq. (32) is intractable in general. To address this, the subsequent analysis will proposed a critic-based learning framework.

5. Design of dynamic event-triggered controls and critic NNS

The event-triggering instants are represented by an increasing sequence $\{t_s\}_{s=0}^\infty$ with $t_0 = 0$. Subsequently, the triggering error is

$$e(t) = \bar{\zeta}(t) - \zeta(t), \quad t \in [t_s, t_{s+1}), \quad (40)$$

with $\bar{\zeta}(t) = \zeta(t_s)$.

Based on (31), the near OC law is formulated as

$$\mu_i^*(\bar{\zeta}) = \rho_{1,i} \tanh \left(-\frac{1}{2\rho_{1,i}} R_{ii}^{-1} E_i^T \nabla \mathcal{V}_i^*(\bar{\zeta}) \right) + \rho_{2,i} \quad (41)$$

Assumption 3 (Wei and Jiang (2025)). There exists a positive definite function $\mathcal{V}$ so that for ζ and e , the following inequality holds:

$$\mathcal{L} \mathcal{V} \leq j \|e\|^2 - \iota \|\zeta\|^2 \quad (42)$$

where $\iota > 0$ and $j > 0$.

5.1. Critic NN design

Based on the NN approximation on properties, the cost function is

$$\mathcal{V}_i^*(\zeta(t)) = \omega_{ci}^T \psi_{ci}(\zeta) + \varepsilon_{ci}, \quad (43)$$

where ω_{ci} is the critic NN weight, $\psi_{ci}(\zeta)$ represents activation functions and ε_{ci} is the critic residual error. The corresponding gradient $\nabla \mathcal{V}_i^*(\zeta)$ is given by

$$\nabla \mathcal{V}_i^*(\zeta(t)) = \nabla \psi_{ci}^T(\zeta) \omega_{ci} + \nabla \varepsilon_{ci}, \quad (44)$$

where $\nabla \psi_{ci}(\zeta) = \frac{\partial \psi_{ci}(\zeta)}{\partial \zeta}$ and $\nabla \varepsilon_{ci} = \frac{\partial \varepsilon_{ci}}{\partial \zeta}$.

Based on (41) and (44), the event-triggered near OC policy is given by

$$\mu_i^*(\bar{\zeta}) = \rho_{1,i} \tanh \left(-\frac{1}{2\rho_{1,i}} R_{ii}^{-1} E_i^T (\nabla \psi_{ci}^T(\bar{\zeta}) \omega_{ci} + \nabla \varepsilon_{ci}) \right) + \rho_{2,i} \quad (45)$$

The estimated value of ω_{ci} is $\hat{\omega}_{ci}$, so the resulting approximate value function is

$$\hat{\mathcal{V}}_i(\zeta(t)) = \hat{\omega}_{ci}^T \psi_{ci}(\zeta). \quad (46)$$

The gradient of $\hat{\mathcal{V}}_i(\zeta(t))$ is

$$\nabla \hat{\mathcal{V}}_i(\zeta(t)) = \nabla \psi_{ci}^T(\zeta) \hat{\omega}_{ci}, \quad (47)$$

In light of (41) and (47), one gets

$$\hat{\mu}_i(\bar{\zeta}) = \rho_{1,i} \tanh \left(-\frac{1}{2\rho_{1,i}} R_{ii}^{-1} E_i^T \nabla \psi_{ci}^T(\bar{\zeta}) \hat{\omega}_{ci} \right) + \rho_{2,i} \quad (48)$$

The Hamiltonian equation is subsequently utilized by replacing $\mu_i^*(\bar{x})$ with $\hat{\mu}_i(\bar{\zeta})$ to derive

$$\begin{aligned} H_i(\zeta, \hat{\mu}_1(\bar{\zeta}), \dots, \hat{\mu}_N(\bar{\zeta}), \hat{\mathcal{V}}_i(\zeta)) \\ &= \hat{\omega}_{ci}^T \nabla \psi_{ci}(\zeta) [f + \sum_{j=1}^N E_j \hat{\mu}_j(\bar{\zeta})] - a \hat{\mathcal{V}}_i(\zeta) + \zeta^T Q_i \zeta \\ &\quad + \frac{1}{2} \text{Tr} \{ h^T \nabla^2 \hat{\mathcal{V}}_i h \} + \sum_{j=1}^N \mathcal{R}_i(\hat{\mu}_j(\bar{\zeta})) + B_r(\zeta) \quad (49) \end{aligned}$$

Define $\xi = \nabla \psi_{ci}(\zeta) [f + \sum_{j=1}^N E_j \hat{\mu}_j(\bar{\zeta})]$, then the Bellman residual becomes

$$\begin{aligned} e_{ci} &= H_i(\zeta, \hat{\mu}_1(\bar{\zeta}), \dots, \hat{\mu}_N(\bar{\zeta}), \hat{\mathcal{V}}_i(\zeta)) - H_i(\zeta, \mu_1^*, \dots, \mu_N^*, \mathcal{V}_i^*(\zeta)) \\ &= \hat{\omega}_{ci}^T \xi - a \hat{\mathcal{V}}_i(\zeta) + \zeta^T Q_i \zeta + \frac{1}{2} \text{Tr} \{ h^T \nabla^2 \hat{\mathcal{V}}_i h \} \\ &\quad + \sum_{j=1}^N \mathcal{R}_i(\hat{\mu}_j(\bar{\zeta})) + B_r(\zeta) \quad (50) \end{aligned}$$

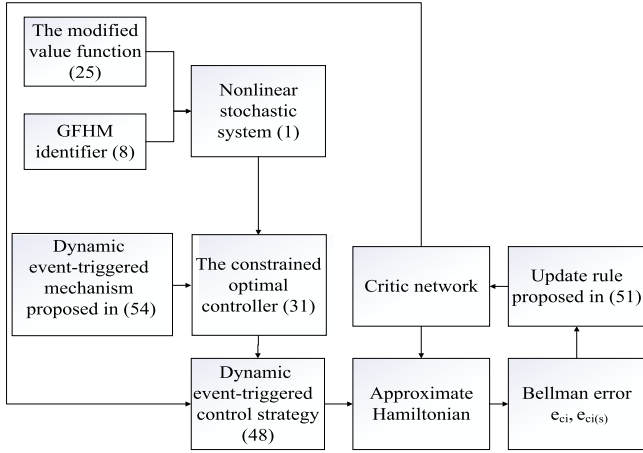


Fig. 1. The architecture of the developed control algorithm.

The critic weight update rule is

$$\dot{\hat{w}}_{ci} = -\frac{\rho_{ci}\xi}{(1+\xi^T\xi)^2}e_{ci} - \sum_{s=1}^{n_c} \frac{\rho_{ci}\xi(s)}{(1+\xi(s)^T\xi(s))^2}e_{ci(s)}, \quad t, t_s \in [t_j, t_{j+1}) \quad (51)$$

with $s \in \{1, 2, \dots, n_c\}$ denoting the data index, $\xi(s) = \xi(t_s)$, $e_{ci(s)} = e_{ci}(t_s)$ and $\hat{w}_{ci} = \omega_{ci} - \hat{w}_{ci}$.

Thus, it follows that

$$\begin{aligned} \dot{\hat{w}}_{ci} = & -\rho_{ci} \left(\psi \psi^T + \sum_{s=1}^{n_c} \psi(s) \psi(s)^T \right) \hat{w}_{ci} + \frac{\rho_{ci} \psi}{1 + \xi^T \xi} \bar{\epsilon}_{ci} \\ & + \sum_{s=1}^{n_c} \frac{\rho_{ci} \psi(s)}{1 + \xi(s)^T \xi(s)} \bar{\epsilon}_{ci(s)} \quad t, t_s \in [t_j, t_{j+1}) \end{aligned} \quad (52)$$

where $\rho_{ci} > 0$, $i = 0, 1, 2, \dots, N$ is the learning rate, $\psi = \xi / (1 + \xi^T \xi)$, $\psi(s) = \xi(s) / (1 + \xi(s)^T \xi(s))$, $\bar{\epsilon}_{ci} = -\nabla \epsilon_{ci}^T (f + \sum_{j=1}^N E_j \hat{\mu}_j)$, $\bar{\epsilon}_{ci(s)} = -\nabla \epsilon_{ci}^T (f(\zeta(t_s)) + \sum_{j=1}^N E_j \hat{\mu}_j(\zeta(t_s)))$.

In order to clarify how the GFHM identifier, constraint controller, and critical network interact with each other, Fig. 1 presents the structural diagram of the proposed control scheme.

5.2. Stability analysis

The DETM adjusts the triggering threshold online, allowing the triggering conditions to change in real-time with the system state, thereby significantly reducing communication load while ensuring control performance.

Assumption 4. The norm of $\bar{\epsilon}_{ci}$ is bounded, such that $\|\bar{\epsilon}_{ci}\| \leq b_{\bar{\epsilon}_{ci}}$, where $b_{\bar{\epsilon}_{ci}}$ is a positive constant. Moreover, the NN fulfills $\|\nabla \phi_{ci}\| \leq b'_{\psi_{ci}}$ and $\|\nabla \epsilon_{ci}\| \leq b'_{\epsilon_{ci}}$, where $b'_{\psi_{ci}}$ and $b'_{\epsilon_{ci}}$ are positive constants.

In the proposed DETC approach, an internal dynamic variable δ_ζ is introduced as

$$\dot{\delta}_\zeta = -\gamma \delta_\zeta + \Gamma(\zeta, e), \quad \delta_\zeta^0 = \delta_\zeta(0), \quad (53)$$

where $\gamma > 0$, $\delta_\zeta^0 > 0$ and $\Gamma(\zeta, e) = (1 - \sigma)\iota\|\zeta\|^2 - J\|e\|^2$.

The following rule defines the DETM:

$$t_{s+1} = \inf \{t \geq t_s \mid \eta \Gamma(\zeta, e) + \delta_\zeta \leq 0\} \quad (54)$$

with $\eta \geq 0$.

Lemma 3. The DETM in (54) guarantees that the solution δ_ζ of the (53) remains strictly positive for all time.

Proof. When $\eta = 0$, (54) directly implies $\delta_\zeta > 0$. It remains to consider $\eta > 0$, where (54) gives $\Gamma(\zeta, e) > -\frac{1}{\eta}\delta_\zeta$, then

$$\dot{\delta}_\zeta = -\gamma \delta_\zeta + \Gamma(\zeta, e) > -\left(\gamma + \frac{1}{\eta}\right)\delta_\zeta \quad (55)$$

From the comparison principle, we deduce that

$$\delta_\zeta \geq \delta_\zeta^0 e^{-(\gamma + \frac{1}{\eta})t} \quad (56)$$

indicating that δ_ζ remains positive provided $\delta_\zeta^0 > 0$. $\square$

Theorem 3. Consider the stochastic system (1) under the control policy (48) and the single-critic NN weight update law in (51). Under the condition that Assumptions 1–4 are all valid, if the DETC in (54) with parameters $\eta \geq 0$ and $\sigma \in (0, 1)$, then: (i) the state variables of the closed-loop system and the weight estimation error of the NN can be kept stable within the UUB; (ii) the system trajectory always remains within the safe region $\mathfrak{D}$.

Proof. Lyapunov function $\mathcal{M}(t)$ is presented as

$$\mathcal{M}(t) = \mathcal{M}_1(t) + \mathcal{M}_2(t) + \mathcal{M}_3(t) + \mathcal{M}_4(t) \quad (57)$$

with

$$\mathcal{M}_1(t) = \sum_{i=1}^N \mathcal{V}_i^*(\zeta(t)), \quad \mathcal{M}_2(t) = \sum_{i=1}^N \mathcal{V}_i^*(\bar{\zeta})$$

$$\mathcal{M}_3(t) = \frac{1}{2} \sum_{i=1}^N \bar{w}_{ci}^T \bar{w}_{ci}, \quad \mathcal{M}_4(t) = \delta_\zeta(t)$$

Case 1: No event has been triggered, i.e., $\forall t \in [t_s, t_{s+1})$, it has $\mathcal{L}\mathcal{M}_2(t) = 0$, $\mathcal{L}\mathcal{M}(t) = \mathcal{L}\mathcal{M}_1(t) + \mathcal{L}\mathcal{M}_3(t) + \mathcal{L}\mathcal{M}_4(t)$.

First, for the function $\mathcal{M}_1(t)$, apply the developed DETC $\hat{\mu}_j(\bar{\zeta})$ in (48), then, it obtains

$$\begin{aligned} \mathcal{L}\mathcal{M}_1(t) = & \sum_{i=1}^N \left[\nabla \mathcal{V}_i^{*T}(\zeta) \left(f(\zeta) + \sum_{j=1}^N E_j \mu_j^*(\bar{\zeta}) \right) + \frac{1}{2} h^T(\zeta) \nabla^2 \mathcal{V}_i^*(\zeta) h(\zeta) \right. \\ & \left. + \nabla \mathcal{V}_i^{*T}(\zeta) \sum_{j=1}^N E_j(\zeta) \left(\hat{\mu}_j(\bar{\zeta}) - \mu_j^*(\bar{\zeta}) \right) \right] \end{aligned} \quad (58)$$

Based on Assumption 3, (58) can be transformed into

$$\begin{aligned} \mathcal{L}\mathcal{M}_1(t) \leq & \sum_{i=1}^N \left[-\iota\|\zeta\|^2 + J\|e\|^2 + \nabla \mathcal{V}_i^{*T}(\zeta) \sum_{j=1}^N E_j(\zeta) \left(\hat{\mu}_j(\bar{\zeta}) - \mu_j^*(\bar{\zeta}) \right) \right] \\ \leq & \sum_{i=1}^N \left[\sum_{j=1}^N \frac{1}{2} \|E_j\|^2 \|\Pi_1\|^2 + \frac{1}{2} \|\nabla \mathcal{V}_i^*(\zeta)\|^2 - \iota\|\zeta\|^2 + J\|e\|^2 \right] \end{aligned} \quad (59)$$

where $\Pi_1 = \hat{\mu}_j(\bar{\zeta}) - \mu_j^*(\bar{\zeta})$.

The Π_1 in (59) can be rephrased as

$$\begin{aligned} \Pi_1 = & \rho_{1,j} \tanh \left(-\frac{1}{2\rho_{1,j}} R_{ij}^{-1} E_j^T (\nabla \psi_{ci}^T(\bar{\zeta}) \hat{w}_{ci}) \right) + \rho_{2,j} \\ & - \rho_{1,j} \tanh \left(-\frac{1}{2\rho_{1,j}} R_{ij}^{-1} E_j^T (\nabla \psi_{ci}^T(\bar{\zeta}) \omega_{ci} + \nabla \epsilon_{ci}) \right) - \rho_{2,j} \\ \leq & \frac{1}{2} \left\| R_{ij}^{-1} E_j^T (\nabla \psi_{ci}^T(\bar{\zeta}) \omega_{ci} + \nabla \epsilon_{ci}) - R_{ij}^{-1} E_j^T (\nabla \psi_{ci}^T(\bar{\zeta}) \hat{w}_{ci}) \right\| \\ = & \frac{1}{2} \left\| R_{ij}^{-1} E_j^T (\nabla \psi_{ci}^T(\bar{\zeta}) \hat{w}_{ci}) + R_{ij}^{-1} E_j^T (\nabla \epsilon_{ci}) \right\| \end{aligned} \quad (60)$$

We deduce from (60) and Assumption 4 that

$$\|\Pi_1\|^2 \leq \frac{1}{2} \Pi_2 \|\bar{W}_{ci}\|^2 + \frac{1}{2} \Pi_2 \|b'_{\epsilon_{ci}}\|^2 \quad (61)$$

where $\Pi_2 = \|R_{ij}^{-1} E_j^T b'_{\psi_{ci}}\|^2$.

Based on (59) and (61), one gets

$$\begin{aligned} \mathcal{LM}_1(t) \leq & \sum_{i=1}^N \left[\frac{1}{2} \|\nabla \mathcal{V}_i^*(\zeta)\|^2 + \sum_{j=1}^N \frac{\|E_j\|^2}{4} \Pi_2 \|\tilde{\omega}_{ci}\|^2 \right. \\ & \left. - t\|\zeta\|^2 + j\|e\|^2 + \sum_{j=1}^N \frac{\|E_j\|^2}{4} \Pi_2 \|b'_{\epsilon_{ci}}\|^2 \right] \end{aligned} \quad (62)$$

Based on (52), the $\mathcal{LM}_3(t)$ yield

$$\begin{aligned} \mathcal{LM}_3(t) = & \sum_{i=1}^N \left[-\rho_{ci} \tilde{\omega}_{ci}^T \left(\psi \psi^T + \sum_{k=1}^{n_c} \psi_{(s)} \psi_{(s)}^T \right) \tilde{\omega}_{ci} \right. \\ & \left. + \frac{\rho_{ci} \tilde{\omega}_{ci}^T \psi}{1 + \xi^T \xi} \tilde{\epsilon}_{ci} + \sum_{s=1}^{n_c} \frac{\rho_{ci} \tilde{\omega}_{ci}^T \psi_{(s)}}{1 + \xi_{(s)}^T \xi_{(s)}} \tilde{\epsilon}_{ci(s)} \right] \end{aligned} \quad (63)$$

Using Young's inequality, it obtains

$$\begin{aligned} \frac{\rho_{ci} \tilde{\omega}_{ci}^T \psi}{1 + \xi^T \xi} \tilde{\epsilon}_{ci} & \leq \frac{\rho_{ci}}{2} \tilde{\omega}_{ci}^T \psi \psi^T \tilde{\omega}_{ci} + \frac{\rho_{ci}}{2} \tilde{\epsilon}_{ci}^T \tilde{\epsilon}_{ci} \\ \sum_{s=1}^{n_c} \frac{\rho_{ci} \tilde{\omega}_{ci}^T \psi_{(s)}}{1 + \xi_{(s)}^T \xi_{(s)}} \tilde{\epsilon}_{ci(s)} & \leq \frac{\rho_{ci}}{2} \sum_{s=1}^{n_c} \tilde{\omega}_{ci}^T \psi_{(s)} \psi_{(s)}^T \tilde{\omega}_{ci} + \frac{\rho_{ci}}{2} \sum_{s=1}^{n_c} \tilde{\epsilon}_{ci(s)}^T \tilde{\epsilon}_{ci(s)} \end{aligned} \quad (64)$$

By combining (64) and Assumption 4, (63) can be transformed into

$$\mathcal{LM}_3(t) \leq \sum_{i=1}^N \left[-\frac{\rho_{ci}}{2} \lambda_{\min}(\Phi(\psi, \psi_{(s)})) \|\tilde{\omega}_{ci}\|^2 + \frac{\rho_{ci}(n_c + 1)}{2} b_{\tilde{\epsilon}_{ci}}^2 \right] \quad (65)$$

where $\Phi(\psi, \psi_{(s)}) = \psi \psi^T + \sum_{s=1}^{n_c} \psi_{(s)} \psi_{(s)}^T$.

From (53), we can get

$$\mathcal{LM}_4(t) = -\gamma \delta_\zeta + \Gamma(\zeta, e) \quad (66)$$

According to (62), (65) and (66), we can infer that

$$\begin{aligned} \mathcal{LM}(t) \leq & -(1 - \theta) t \|\zeta\|^2 + j \|e\|^2 - \sum_{i=1}^N \left[\frac{\rho_{ci}}{2} \lambda_{\min}(\Phi(\psi, \psi_{(s)})) \right. \\ & \left. - \sum_{j=1}^N \frac{\|E_j\|^2}{4} \Pi_2 \right] \|\tilde{\omega}_{ci}\|^2 - \sigma t \|\zeta\|^2 + \Psi + -\gamma \delta_\zeta + \Gamma(\zeta, e) \end{aligned} \quad (67)$$

where $0 < \sigma < 1$, $\Psi = \sum_{i=1}^N \left[\sum_{j=1}^N \frac{\|E_j\|^2}{4} \Pi_2 \|b'_{\epsilon_{ci}}\|^2 + \frac{\rho_{ci}(n_c + 1)}{2} b_{\tilde{\epsilon}_{ci}}^2 + \frac{1}{2} \|\nabla \mathcal{V}_i^*(\zeta)\|^2 \right]$.

Therefore, only the following conditions need to be met

$$\|\tilde{\omega}_{ci}\| \geq \sqrt{\frac{\Psi}{\sum_{i=1}^N \left[\frac{\rho_{ci}}{2} \lambda_{\min}(\Phi(\psi, \psi_{(s)})) - \sum_{j=1}^N \frac{\|E_j\|^2}{4} \Pi_2 \right]}} \quad (68)$$

and $\Gamma(\zeta, e) < 0$ in (53), then

$$\mathcal{LM}(t) \leq -(\sigma t \|\zeta\|^2 + \gamma \delta_\zeta) \leq 0. \quad (69)$$

Case 2: At the event-triggering instant $t = t_{s+1}$, $\Delta \mathcal{M}$ reduces to

$$\begin{aligned} \Delta \mathcal{M} = & \mathcal{M}(\zeta(t_{s+1})) - \mathcal{M}(\zeta(t_{s+1}^-)) \\ = & \Delta \mathcal{M}_1 + \Delta \mathcal{M}_2 + \Delta \mathcal{M}_3 + \Delta \mathcal{M}_4 \end{aligned} \quad (70)$$

where

$$\begin{aligned} \Delta \mathcal{M}_1 = & \mathcal{V}_i^*(\zeta(t_{s+1}^+)) - \mathcal{V}_i^*(\zeta(t_{s+1}^-)), \\ \Delta \mathcal{M}_2 = & \mathcal{V}_i^*(\tilde{\zeta}(t_{s+1})) - \mathcal{V}_i^*(\tilde{\zeta}(t_s)), \\ \Delta \mathcal{M}_3 = & \frac{1}{2} \tilde{\omega}_{ci}^T(t_{s+1}^+) \tilde{\omega}_{ci}(t_{s+1}^+) - \frac{1}{2} \tilde{\omega}_{ci}^T(t_{s+1}) \tilde{\omega}_{ci}(t_{s+1}), \\ \Delta \mathcal{M}_4 = & \delta_\zeta(t_{s+1}^+) - \delta_\zeta(t_{s+1}). \end{aligned}$$

Based on case 1, $\mathcal{LM}(t)$ exhibits strict monotonic decrease over the interval $[t_s, t_{s+1})$. Thus, one gets

$$\mathcal{LM}(t_{s+1} + e) < \mathcal{LM}(t_{s+1}), \forall e \in (0, t_{s+1} - t_s) \quad (71)$$

Take the limit on both sides of (71) to get $\mathcal{LM}(t_{s+1}^+) = \lim_{e \rightarrow 0^+} \mathcal{LM}(t_{s+1} + e)$, that is $\mathcal{LM}(t_{s+1}^+) \leq \mathcal{LM}(t_{s+1})$. In this case, we can obtain

$$\begin{aligned} \mathcal{V}_i^*(\zeta(t_{s+1}^+)) + \frac{1}{2} \tilde{\omega}_{ci}^T(t_{s+1}^+) \tilde{\omega}_{ci}(t_{s+1}^+) + \delta_\zeta(t_{s+1}^+) \\ \leq \mathcal{V}_i^*(\zeta(t_{s+1})) + \frac{1}{2} \tilde{\omega}_{ci}^T(t_{s+1}) \tilde{\omega}_{ci}(t_{s+1}) + \delta_\zeta(t_{s+1}) \end{aligned} \quad (72)$$

In addition, $\mathcal{V}_i^*$ exhibits continuous differentiability at the event-triggering time t_{s+1} for $s + 1 \in N$. Invoking the results from case 1 yields

$$\mathcal{V}_i^*(\zeta(t_{s+1}^+)) \leq \mathcal{V}_i^*(\zeta(t_{s+1})) \quad (73)$$

Considering (72) and (73), one gets $\Delta \mathcal{M} \leq 0$. This completes the proof. $\square$

Remark 5. According to the event-triggering mechanism, the control strategy is only updated when the event-triggering mechanism is met. Therefore, the stability proof is explained in two parts: on the one hand, to demonstrate stability upon event-triggering, on the other hand, stability is demonstrated between the triggering moments of two events.

6. Simulation results

6.1. Example 1

Consider the stochastic system with asymmetric input saturations:

$$d\zeta = \left[f(\zeta) + \sum_{j=1}^3 E_j \mu_j \right] dt + h(\zeta) dw, \quad (74)$$

with

$$\begin{aligned} f(\zeta) = & \begin{bmatrix} \zeta_2 - 2\zeta_1 \\ -7 \sin(\zeta_1) - 3.5\zeta_2 - \zeta_1 \end{bmatrix}, \\ E_1 = & \begin{bmatrix} 0 \\ \sin(\zeta_1) \end{bmatrix}, \quad E_2 = \begin{bmatrix} 0 \\ \cos(\zeta_1) \end{bmatrix}, \\ E_3 = & \begin{bmatrix} 0 \\ 0.5 \cos(\zeta_1) \sin(\zeta_1) \end{bmatrix}, \quad h(\zeta) = \begin{bmatrix} 0 \\ 0.1 \sin(\zeta_2) \end{bmatrix}. \end{aligned}$$

with $\zeta = [\zeta_1, \zeta_2]^T \in \mathbb{R}^2$ being the state vector and the initial value $\zeta_0 = [1, -1]^T$. The bounds of the control inputs μ_1, μ_2 and μ_3 are set as $\mu_{1,\min} = 0.2, \mu_{2,\min} = -0.3, \mu_{3,\min} = 0.8$ and $\mu_{1,\max} = 1, \mu_{2,\max} = 0.4, \mu_{3,\max} = 1$, respectively, therefore $\rho_{1,1} = 0.4, \rho_{1,2} = 0.35, \rho_{1,3} = 0.1$ and $\rho_{2,1} = 0.6, \rho_{2,2} = 0.05, \rho_{2,3} = 0.9$. The value function parameters are chosen as $Q_1 = 3I_2, Q_2 = 3I_2, Q_3 = 2I_2$.

This paper takes the nonlinear function $f(\zeta)$ as the unknown term. According to (8), the following GFHM-based identifier is utilized to approximate the unavailable terms:

$$\dot{\hat{\zeta}} = \hat{I} + \hat{\Xi}^T \tanh(\zeta(\hat{X}_s - \hat{C})) + D\hat{\zeta} + \sum_{j=1}^N \hat{E}_j \mu_j.$$

where $\hat{X}_s = \Phi(\hat{\zeta} \otimes 1_8), \hat{\Xi} = [\hat{\Xi}_1, 0_8; 0_8, \hat{\Xi}_2], \hat{C} = [\hat{C}_1, \hat{C}_2]$. The system state dimension is $n = 2$, and the number of hyperbolic tangent basis functions for each state is $k = 8, D = \text{diag}[-20, -25]$. The GFHM identifier adopts the following parameters: the scalar learning rates are set $\alpha_1 = 3.5$ and $\alpha_3 = 1.5$. The learning rate matrices for B parameters are designed as $\alpha_{2,v} = 3I_8 \in \mathbb{R}^{8 \times 8}, v = 1, 2$. The learning rate matrices for E parameters are set as $\alpha_{4,j} = 3.5I_2 \in \mathbb{R}^{2 \times 2}, j = 1, 2, 3$. And the $\beta_1 = \beta_2 = \beta_3 = 0.0005$.

Furthermore, the unsafe region is defined as a circular area centered at $(0.12, -0.8)$ with radius $r = 0.08$, and the damping factor is set as $\rho = 10$. Then, the $\mathfrak{B}(\zeta)$ design as $\text{atan}\left(1/\left(\sqrt{(\zeta_1 - 0.12)^2 + (\zeta_2 + 0.8)^2} - 0.08\right)\right)$. The $S(\zeta)$ is chosen as

$\sqrt{(\zeta_1 - 0.12)^2 + (\zeta_2 + 0.8)^2} - 0.08$, where $\mathfrak{B}_r(\zeta) = \rho \frac{\mathfrak{B}(\zeta)}{S(\zeta)}$ serves as the CBF that penalizes unsafe behaviors near the danger zone. In addition, $T = 0.1s, \rho_{c1} = 0.5, \rho_{c2} = 0.6, \rho_{c3} = 0.6$ are the learning rates. Activation

functions are selected as $\psi_{ci}(\zeta) = [\zeta_1^2, \zeta_1 \zeta_2, \zeta_2^2, \zeta_1^3, \zeta_2^3]^\top$. The three sets of estimated weights are denoted as $\hat{\omega}_{c1} = [\omega_{c1}^1, \omega_{c1}^2, \omega_{c1}^3, \omega_{c1}^4, \omega_{c1}^5]^\top$, $\hat{\omega}_{c2} = [\omega_{c2}^1, \omega_{c2}^2, \omega_{c2}^3, \omega_{c2}^4, \omega_{c2}^5]^\top$, and $\hat{\omega}_{c3} = [\omega_{c3}^1, \omega_{c3}^2, \omega_{c3}^3, \omega_{c3}^4, \omega_{c3}^5]^\top$, respectively.

As illustrated in Fig. 2, the proposed GFHM identifier can satisfactorily approximate the composite nonlinear term $g = f(\zeta) + \frac{3}{4}\|h\|^4\zeta$. This performance validates the correctness of the theoretical analysis of Theorem 1. In addition, the recognizer successfully reconstructed the unknown dynamics $f(\zeta)$, thereby transforming the unknown system information into usable information, laying the foundation for the implementation of subsequent controllers.

Fig. 3 presents the time evolution of the system states. It is observed that both state variables converge to and remain within a sufficiently small neighborhood of the origin. This convergence behavior validates that the proposed control strategy successfully achieves the stabilization objective for system (1). Unlike Zhang et al. (2025b) and Wei et al. (2023), Fig. 4 illustrate the evolution profiles of the near OC policies with AICs. According to the results, it can be seen that the control input remains within the specified asymmetric range and eventually converges to $\rho_{2,j}$.

Figs. 5–7 characterize the convergence trajectories of estimated network weights $\hat{\omega}_{c1}$, $\hat{\omega}_{c2}$ and $\hat{\omega}_{c3}$, respectively. Consequently, the estimated weights converge to $\hat{\omega}_{c1} = [0.6275, -0.0209, 0.7374, 0.0196, -1.2873]^\top$, $\hat{\omega}_{c2} = [0.5942, -0.0946, 0.6612, 0.0221, -1.4588]^\top$, $\hat{\omega}_{c3} = [0.4437, -0.3091, 0.4680, 0.0437, -1.3799]^\top$. The persistent excitation (PE) condition is deactivated at $t = 40$ s, where the PE signal is defined as $3 \exp(-0.015t)(\sin^5(t) + \cos(0.1t)\sin^2(2t) + \cos(t)\sin^2(t) + 1.2\sin^2(-1.2t)\cos(0.5t))$. Fig. 8 describes the evolution process of triggering conditions and event thresholds. Fig. 9 shows the details of the sampling process. Compared to the 800 state samples required by time-triggered control, the event-triggered approach uses only 385 samples, achieving a reduction in controller updates of approximately 51.88%.

Fig. 10 illustrates the danger zone represented by a circular region centered at $(0.14, -0.8)$ with radius 0.1. The results show that: (i) without CBF, the trajectory enters the unsafe region, whereas with CBF, the system successfully avoids it; (ii) when approaching the danger zone, the synergistic effect between CBF and the cost function accelerates the convergence rate of ζ_2 , since the control input directly affects ζ_2 dynamics through the input matrix structure. Consequently, the trajectory bypassing the danger region is achieved, and both states ultimately converge to zero.

6.2. Example 2

Consider the stochastic dynamics of an inverted pendulum, where ω denotes multiplicative white noise (Zhang et al., 2012):

$$ml^2\ddot{\theta} + k\theta + (0.001\varepsilon_t + \hbar)\dot{\theta} - mgl\sin(\theta) = \mu + 2v \quad (75)$$

with the spring coefficient being $k = 10$ N/m, $\hbar = 0.8$ is damping coefficient, the length and the mass are $l = 0.7$ m and $m = 0.5$ kg respectively, the inclination angle of the pendulum is θ , $g = 9.8$ m/s² is the gravitational coefficient.

The Itô formulation of the state-space model is

$$d\zeta(t) = \begin{bmatrix} 0 & \frac{\zeta_2}{ml^2} \\ mgl\sin(\zeta_1) - k\zeta_1 & -\frac{\hbar\zeta_2}{ml^2} \end{bmatrix} dt + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \mu_1 dt + \begin{bmatrix} 0 \\ 2 \end{bmatrix} \mu_2 dt + \begin{bmatrix} 0 \\ -\frac{0.001\zeta_2}{ml^2} \end{bmatrix} d\omega(t) \quad (76)$$

where $\zeta = [\zeta_1, \zeta_2]^\top \in \mathbb{R}^2$ is the state vector and $\zeta_0 = [0.5, 0.1]^\top$. The control inputs bounds for μ_1 and μ_2 are set as $\mu_{1,\min} = -1.3$, $\mu_{2,\min} = 0.1$, $\mu_{1,\max} = -0.6$ and $\mu_{2,\max} = 0.9$, respectively, therefore $\rho_{1,1} = 0.35$, $\rho_{2,1} = -0.95$ and $\rho_{1,2} = 0.4$, $\rho_{2,2} = 0.5$. The value function parameters are chosen

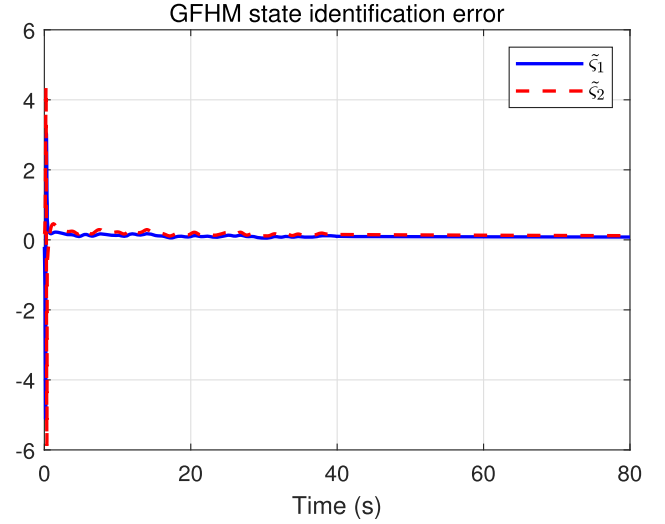


Fig. 2. GFHM states identifier errors in example 1.

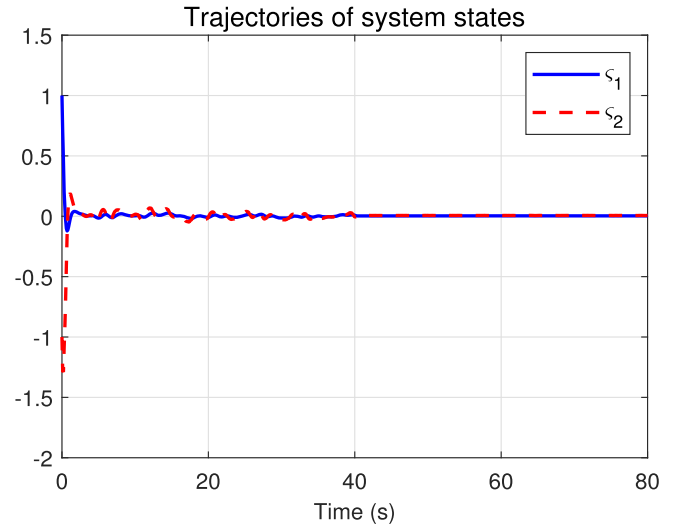


Fig. 3. Curves of system states of system (74).

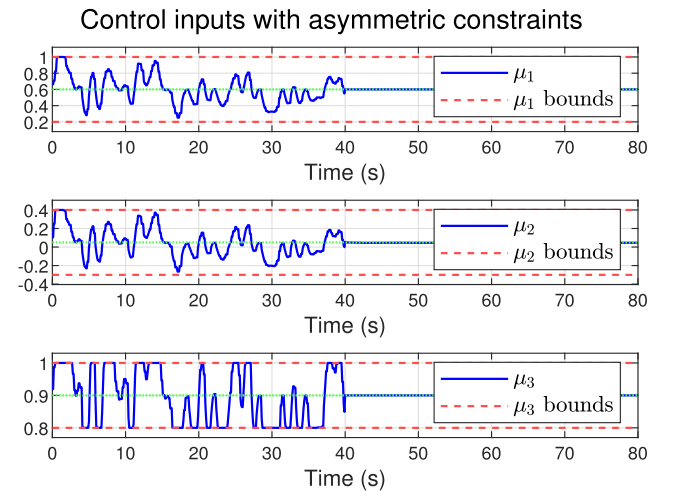


Fig. 4. Control inputs with constraints.

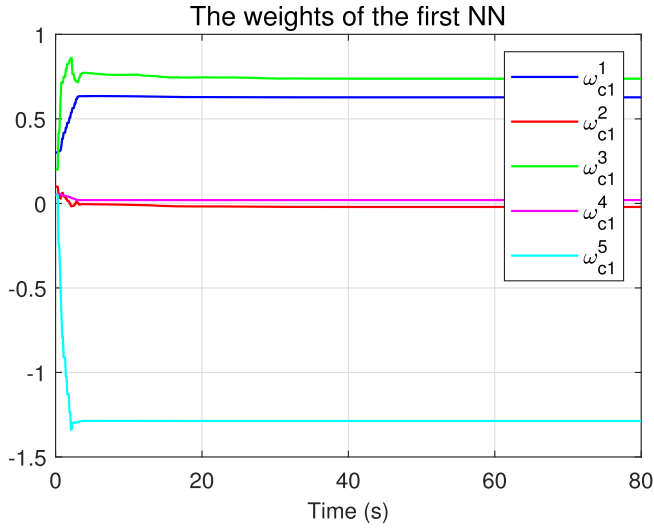


Fig. 5. The weight curves of the first NN in example 1.

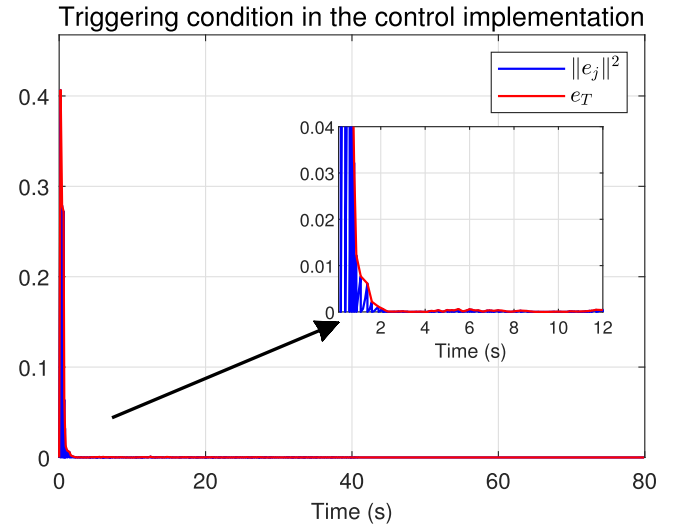
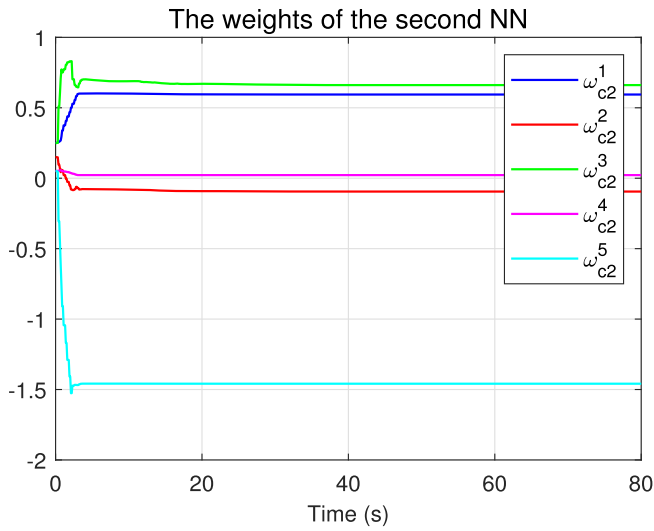
Fig. 8. Triggering condition with $\|e_j\|^2$ and e_T in example 1.

Fig. 6. The weight curves of the second NN in example 1.

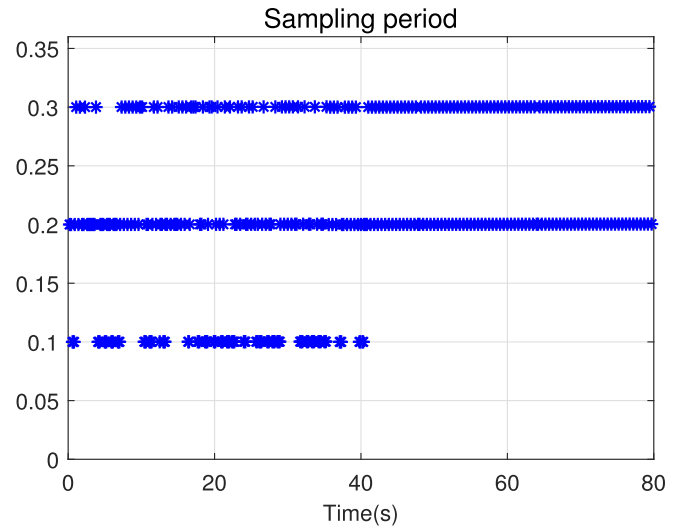


Fig. 9. Sampling period of the learning stage.

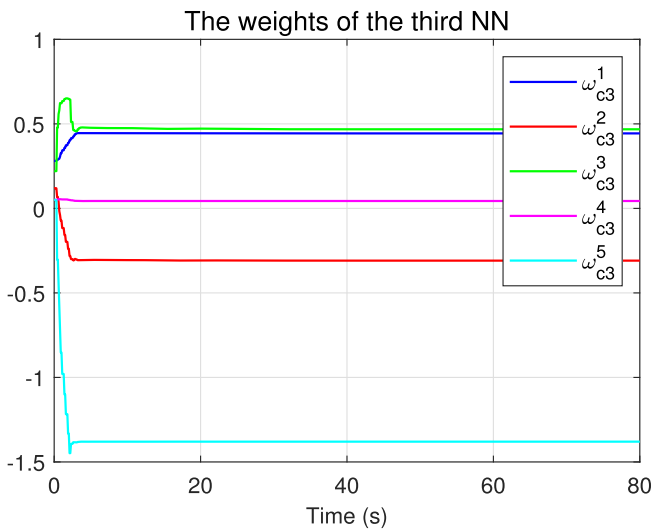


Fig. 7. The weight curves of the third NN in example 1.

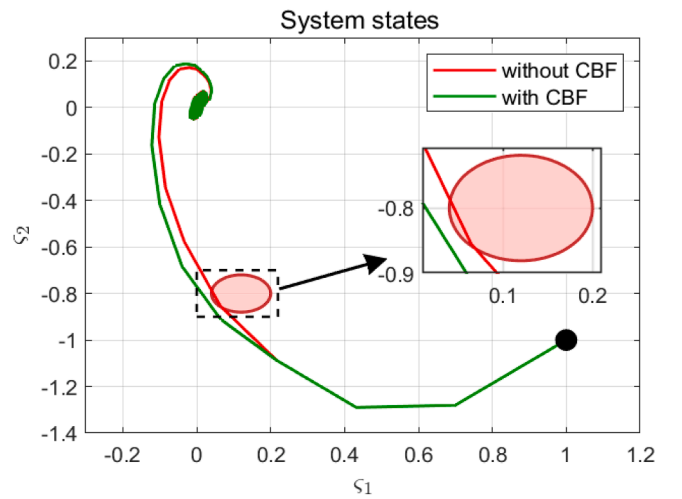


Fig. 10. Comparison of safe and unsafe state trajectories.

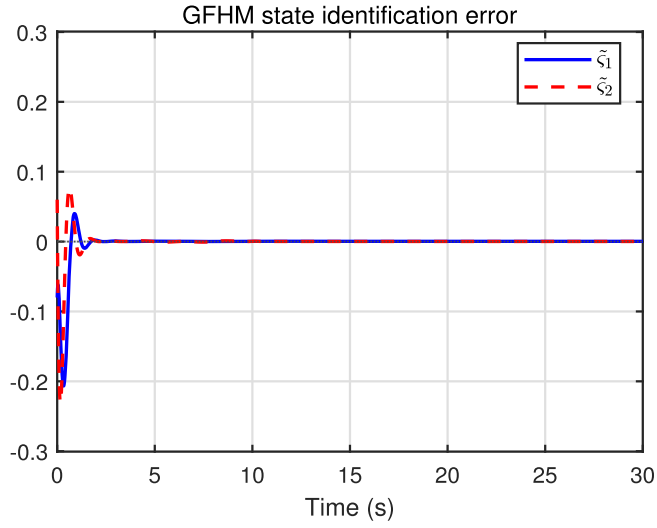


Fig. 11. GFHM states identifier errors in example 2.

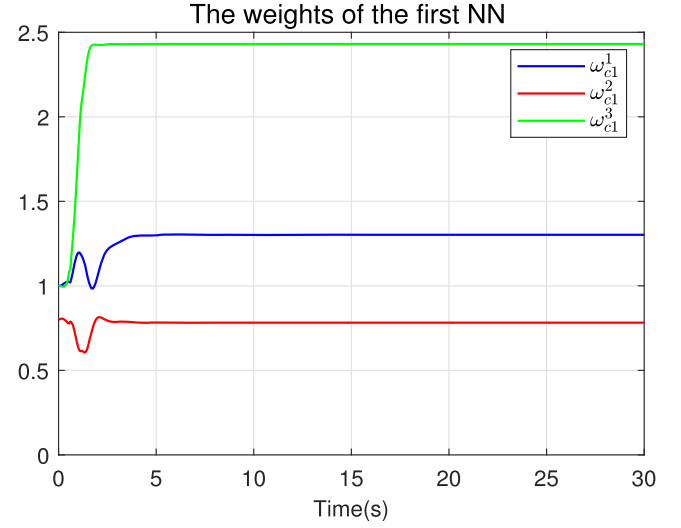


Fig. 14. The weight curves of the first NN in example 2.

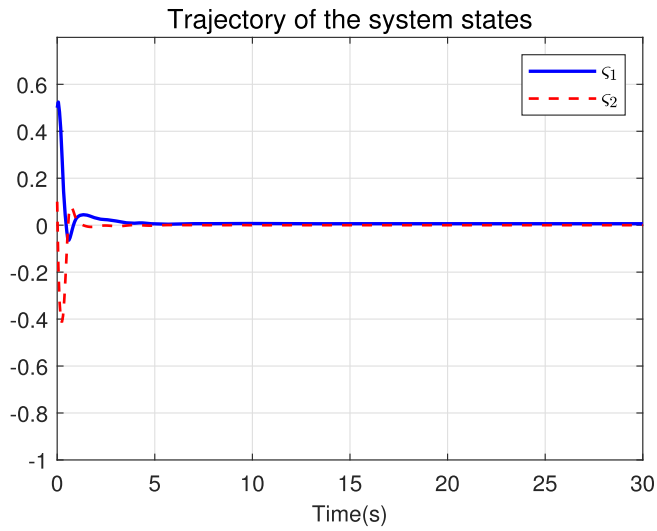


Fig. 12. Curves of system states of system (76).

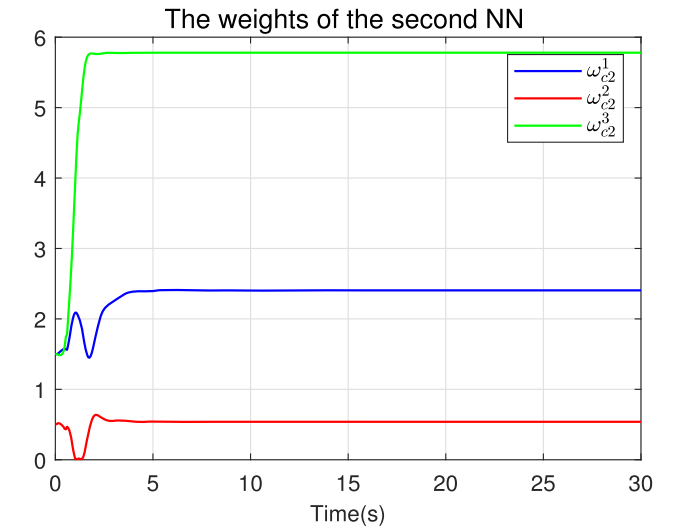


Fig. 15. The weight curves of the second NN in example 2.

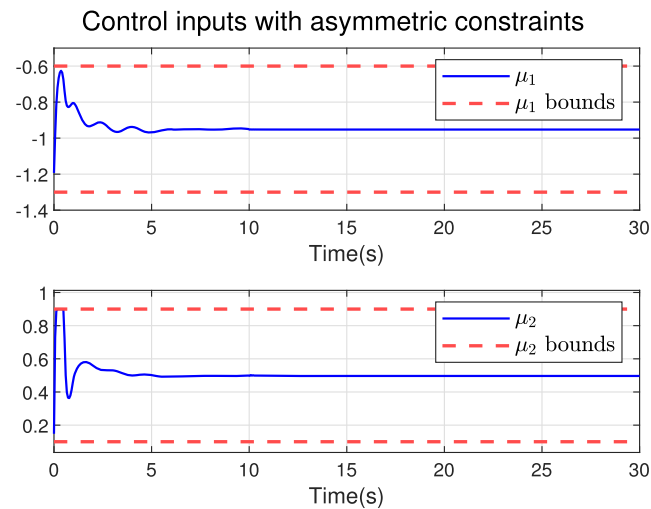
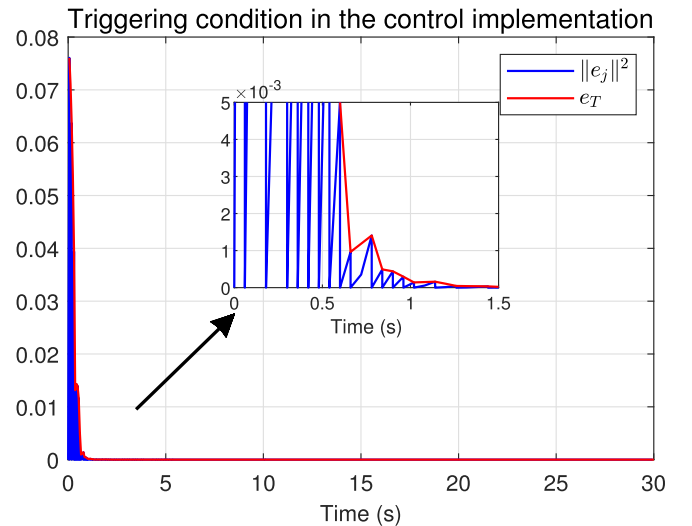


Fig. 13. Control inputs with constraints.

Fig. 16. Triggering condition with $\|e\|^2$ and e_T in example 2.

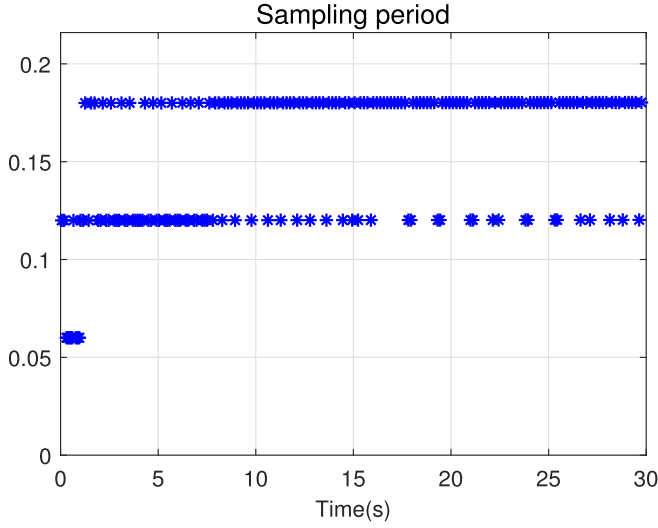


Fig. 17. Sampling period of the learning stage.

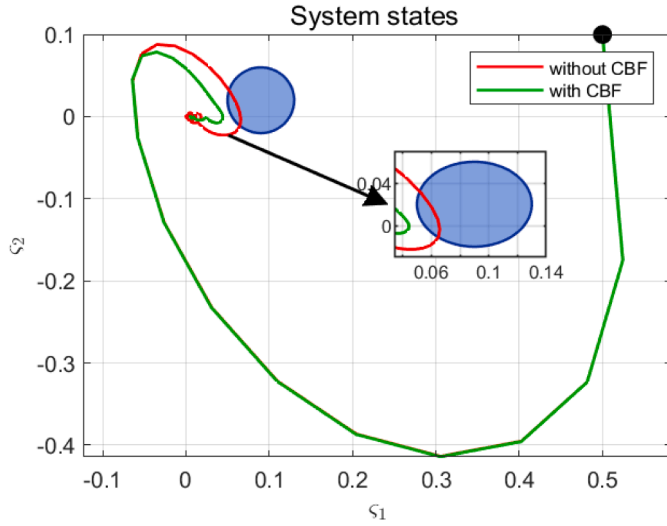


Fig. 18. Comparison of safe and unsafe state trajectories.

as $Q_1 = 1.5I_2$, $Q_2 = 4.5I_2$, and $T = 0.06s$. Activation functions selected as $\psi_{ci}(\varsigma) = [\varsigma_1^2, \varsigma_1\varsigma_2, \varsigma_2^2]^T$ and the learning rates are chosen as $\rho_{c1} = 0.01$, $\rho_{c2} = 0.03$.

The learning rates in (10)–(12) are set as: $\alpha_1 = 0.2$ and $\alpha_3 = 0.1$, the learning rate matrices for $\hat{\Xi}$ parameters are $\alpha_{2,v} = 0.15I_4 \in \mathbb{R}^{4 \times 4}$, $v = 1, 2$, the learning rate matrices for $\hat{E}$ parameters are $\alpha_{4,j} = 0.3I_2 \in \mathbb{R}^{2 \times 2}$, $j = 1, 2$, and $\beta = 0.05$.

The unsafe region is characterized by a circular domain with center coordinates (0.09, 0.02) and radius $r = 0.04$. The damping factor is specified as $\rho = 18$. Accordingly, by Assumption 2, it is known that $B_r(\varsigma) = \rho \frac{\mathfrak{g}(\varsigma)}{S(\varsigma)}$, the $\mathfrak{g}(\varsigma)$ is $\text{atan}\left(1/\left(\sqrt{(\varsigma_1 - 0.09)^2 + (\varsigma_2 - 0.02)^2} - 0.04\right)\right)$. The

$S(\varsigma)$ is chosen as $\sqrt{(\varsigma_1 - 0.09)^2 + (\varsigma_2 - 0.02)^2} - 0.04$. The convergence curve of the recognizer error is displayed in Fig. 11. Fig. 12 shows the evolution of the system state. Unlike the symmetric constraints of Si et al. (2024) and Ming et al. (2022), the evolutions of the control signal in this paper are shown in Fig. 13, indicating that the control input is uniformly and strictly constrained within an asymmetric constraint range, ultimately converging to -0.95 and 0.35 , respectively. From Figs. 14 and 15, it can be seen that the evaluation network weights

$\hat{\omega}_{c1}$ and $\hat{\omega}_{c2}$ will eventually converge to $[1.1557, 1.0225, 2.1559]^T$ and $[1.9663, 1.1729, 4.9565]^T$, respectively.

Fig. 16 illustrates the relationship between triggering conditions and event-triggered thresholds, indicating that the triggering error value is always less than the threshold. The event intervals under the proposed DETC is depicted in Fig. 17. The event-triggered control mechanism only utilizes 196 state samples, whereas the time-triggered control process employs in Zhang et al. (2025d) 500 state samples. Evidently, the event-triggered scheme reduces the controller update rate by up to 60.8%. Fig. 18 presents a comparative analysis of system trajectories with and without CBF, where the circle represents the unsafe region. The results demonstrate a distinct contrast: the CBF-disabled trajectory violates safety by intruding into the dangerous area. This confirms the critical role of CBF in maintaining safe operation.

7. Conclusion

This paper has proposed a safe OC scheme for nonlinear stochastic systems under AICs utilizing RL. The GFHM-based identifier has been developed to approximate unknown system dynamics, with the identification error being proven to converge asymptotically. A nonquadratic cost function incorporating CBF has been designed to handle asymmetric constraints and safety requirements simultaneously. The adoption of the single-critic NN has reduced the computational burden. Additionally, the proposed DETM guarantees a positive minimum inter-event time, which ensures efficient utilization of communication resources. The state and weight have been proven to be UUB through the Lyapunov function while maintaining system safety. This paper provides two examples to verify the effectiveness of the proposed method.

CRediT authorship contribution statement

Yuling Liang: Methodology; **Feng Qin:** Writing – original draft; **Lei Liu:** Investigation; **Yi Zhang:** Supervision; **Zhongyang Ming:** Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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